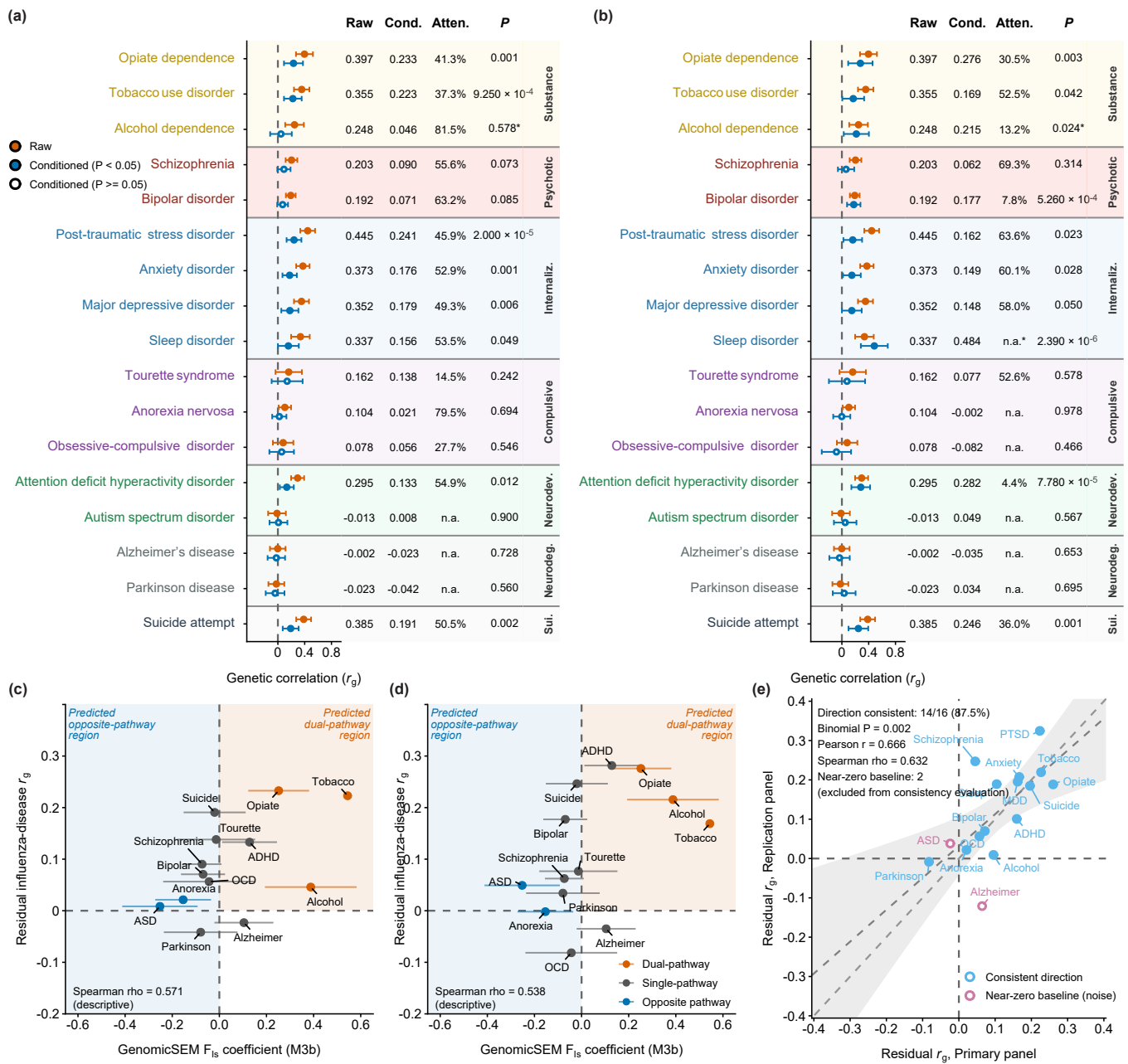

(a)
Raw
Conditioned (P < 0.05)
Conditioned (P >= 0.05)
Raw Cond. Atten. P
Opiate dependence 0.397 0.233 41.3% 0.001
Tobacco use disorder 0.355 0.223 37.3% 9.250 × 10-4
Alcohol dependence 0.248 0.046 81.5% 0.578*
Schizophrenia 0.203 0.090 55.6% 0.073
Bipolar disorder 0.192 0.071 63.2% 0.085
Post-traumatic stress disorder 0.445 0.241 45.9% 2.000 × 10-5
Anxiety disorder 0.373 0.176 52.9% 0.001
Major depressive disorder 0.352 0.179 49.3% 0.006
Sleep disorder 0.337 0.156 53.5% 0.049
Tourette syndrome 0.162 0.138 14.5% 0.242
Anorexia nervosa 0.104 0.021 79.5% 0.694
Obsessive-compulsive disorder 0.078 0.056 27.7% 0.546
Attention deficit hyperactivity disorder 0.295 0.133 54.9% 0.012
Autism spectrum disorder -0.013 0.008 n.a. 0.900
Alzheimer's disease -0.002 -0.023 n.a. 0.728
Parkinson disease -0.023 -0.042 n.a. 0.560
Suicide attempt 0.385 0.191 50.5% 0.002
Substance
Psychotic
Internaliz.
Compulsive
Neurodev.
Neurodeg.
Sui.
0 0.4 0.8
Genetic correlation (rg)
(b)
Raw Cond. Atten. P
Opiate dependence 0.397 0.276 30.5% 0.003
Tobacco use disorder 0.355 0.169 52.5% 0.042
Alcohol dependence 0.248 0.215 13.2% 0.024*
Schizophrenia 0.203 0.062 69.3% 0.314
Bipolar disorder 0.192 0.177 7.8% 5.260 × 10-4
Post-traumatic stress disorder 0.445 0.162 63.6% 0.023
Anxiety disorder 0.373 0.149 60.1% 0.028
Major depressive disorder 0.352 0.148 58.0% 0.050
Sleep disorder 0.337 0.484 n.a.* 2.390 × 10-6
Tourette syndrome 0.162 0.077 52.6% 0.578
Anorexia nervosa 0.104 -0.002 n.a. 0.978
Obsessive-compulsive disorder 0.078 -0.082 n.a. 0.466
Attention deficit hyperactivity disorder 0.295 0.282 4.4% 7.780 × 10-5
Autism spectrum disorder -0.013 0.049 n.a. 0.567
Alzheimer's disease -0.002 -0.035 n.a. 0.653
Parkinson disease -0.023 0.034 n.a. 0.695
Suicide attempt 0.385 0.246 36.0% 0.001
Substance
Psychotic
Internaliz.
Compulsive
Neurodev.
Neurodeg.
Sui.
0 0.4 0.8
Genetic correlation (rg)
(c)
Predicted opposite-pathway region
Predicted dual-pathway region
Tobacco
Suicide
Opiate
Tourette
Schizophrenia
ADHD
Bipolar
OCD
Anorexia
Alcohol
ASD
Parkinson
Alzheimer
Spearman rho = 0.571
(descriptive)
Residual influenza-disease rg
GenomicSEM Fls coefficient (M3b)
(d)
Predicted opposite-pathway region
Predicted dual-pathway region
ADHD
Opiate
Suicide
Alcohol
Bipolar
Tobacco
Schizophrenia
Tourette
ASD
Parkinson
Anorexia
Alzheimer
OCD
Dual-pathway
Single-pathway
Opposite pathway
Spearman rho = 0.538
(descriptive)
Residual influenza-disease rg
GenomicSEM Fls coefficient (M3b)
(e)
Direction consistent: 14/16 (87.5%)
Binomial P = 0.002
Pearson r = 0.666
Spearman rho = 0.632
Near-zero baseline: 2
(excluded from consistency evaluation)
Schizophrenia
PTSD
Anxiety
Tobacco
Opiate
MDD
Suicide
Bipolar
ADHD
ASD
OCD
Anorexia
Alcohol
Parkinson
Alzheimer
Consistent direction
Near-zero baseline (noise)
Residual rg, Replication panel
Residual rg, Primary panel